

Design Before You Build — Paper B

MEDIUM • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working: method marks are awarded even when the final answer is wrong.
- No calculator and no computer. Every number on this paper works by hand.
- Where a question says **sketch**, draw plain boxes and labels only. Colour, real pictures and decoration earn nothing on a wireframe.
- Write **px** on every screen measurement. Give a contrast ratio as a number.
- The mark for each question is shown in square brackets on the right.
- Reminder: **contrast ratio** = $(L_1 + 0.05) \div (L_2 + 0.05)$, where L is relative luminance and L_1 belongs to the **lighter** colour. Ordinary text needs a ratio of at least **4.5**. Large text (24 px or bigger, or 18.66 px bold) needs at least **3**.

SECTION 1 – PROCESS AND VOCABULARY

1. A team is designing a train-ticket app. Ishaan draws the design process like this and says: “That is the double diamond.”

Ishaan's drawing



(a) Give two things his drawing leaves out that the real double diamond shows. [2]

(b) In the box, sketch the real double diamond. Name the four phases, and draw an arrow to show where a failed test can send the team. [2]

SKETCH HERE

2. Three lines from the train-ticket team’s notebook. Write **Define** or **Develop** beside each. [3]

Line from the notebook	Define or Develop?
Sketch six different ways to show a platform change	
Write: “Commuters miss trains because platform changes appear only on a distant board”	
Group forty interview notes into three themes, then pick one	

Which lines are about the problem, and which are about the answer?

3. A student says: “All three of these fixes are UI, because you can see all of them on the screen.”

- (i) Replace the three-screen seat choice with one seat-map screen.
- (ii) Change the Book button to the same blue as the other buttons.
- (iii) Show a spinner and “Booking your seat” after the tap.

- (a) State which one fix is truly a UI fix. [1]

- (b) Explain why each of the other two is a UX fix. [2]

SECTION 2 – MEASURING SCREENS

4. Two button labels from the app, both 16 px and regular weight. L is relative luminance.

Button 1



text #153696 L = 0.05
back #C09C09 L = 0.35

Button 2



text #FFFFFF L = 1.00
back #097E09 L = 0.15

- (a) Calculate the contrast ratio for button 1. [1]

(b) Calculate the contrast ratio for button 2. [1]

(c) State which button passes and which fails for these labels, and give the number each ratio is compared with. [2]

(d) The designer says: “Button 1 is near enough to 4.5.” Explain why that is not good enough. [1]

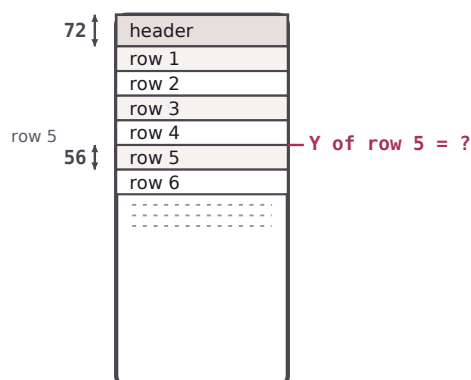
5. A toolbar has **328 px** of usable width. Every button is **44 px** wide, and the gap between neighbouring buttons is **12 px**.

(a) Calculate the total width taken by 6 buttons and the gaps between them. [2]

(b) Can a seventh button fit? Show the width it would need. [1]

n buttons have $n - 1$ gaps between them.

6. A frame is **390 px wide and 844 px high**. A header **72 px** high sits at the top, at $Y = 0$. Below it are list rows, each **56 px** high, with no gaps. Row 1 starts at $Y = 72$.



FIGMA PANEL

Frame: W 390 H 844

Header: H 72, at $Y = 0$

Every row: H 56, no gaps

Row 1: top edge at $Y = 72$

Rows run down to the bottom edge of the frame ($Y = 844$).

(a) Calculate the Y position of row 5. [2]

(b) How many whole rows fit between the header and the bottom of the frame? [1]

SECTION 3 – USERS AND WIREFRAMES

- 7.** Meet the persona. **Mrs Savitri Rao, 66**, books train tickets for her family. She uses her son’s old phone with a cracked screen, her mobile data is slow, and she types with one finger. Her favourite colour is yellow. Her journey booking a ticket is scored from 1 (miserable) to 5 (delighted).

Step	1 Open the app	2 Search trains	3 Choose a seat	4 Passenger details	5 Pay by UPI	6 Get the ticket
Feeling	5	4	2	1	4	5

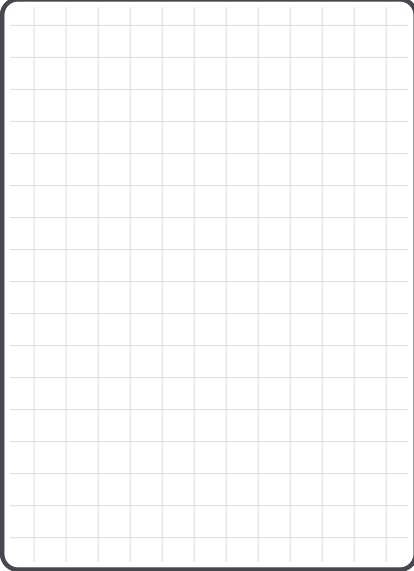
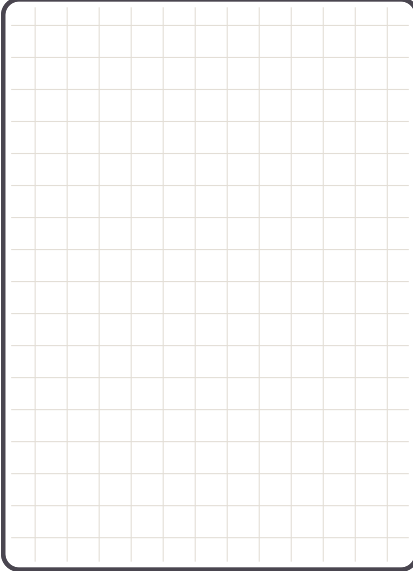
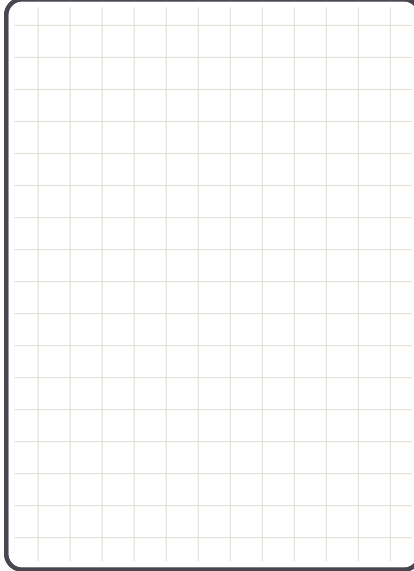
(a) State which step is the lowest point. [1]

(b) Calculate the average feeling score across the six steps. [1]

(c) Give one UX fix aimed at the lowest point. [1]

(d) Copy one detail from Savitri’s card that earns its place on a persona, and say what design decision it changes. [1]

8. The Railway Club needs a three-page website: **Home, About and Contact**. In the [5] frames below, sketch a low-fidelity wireframe for all three pages. It must show: the **same navigation in the same place** on every page; exactly **one heading** per page, before the content; a **button** on Home after its heading; a form with **two fields and a button** on Contact only; and a **footer** that is the last thing on every page. Use the notes lines for anything a box cannot say.

HOME	ABOUT	CONTACT
390 × 844 frame	390 × 844 frame	390 × 844 frame
		
NOTES	NOTES	NOTES

Grey boxes and labels. If you reach for a coloured pencil, put it down.

End of paper. Check that every screen measurement carries px, and that every UI or UX answer says why, not just which.